Experiment: 8

Aim

To create a simple socket program in Python for exchanging basic information between a server and a client.

Theory:

What is Socket Programming?

A **socket** is an endpoint used for communication between two computers or two programs over a network.

Socket programming allows a **client** and a **server** to communicate with each other by sending and receiving data.

Client-Server Model

Socket communication generally follows the **client-server model**.

Server

The server:

1. Creates a socket.
2. Assigns an IP address and port.
3. Waits for a client.
4. Accepts the connection.
5. Receives and sends information.
6. Closes the connection.

Client

The client:

1. Creates a socket.
2. Connects to the server.
3. Sends information.
4. Receives the server's response.
5. Closes the connection.

TCP Socket

In this practical, we use **TCP**.

TCP stands for **Transmission Control Protocol**.

TCP provides:

- Reliable communication
- Connection-oriented communication
- Ordered delivery of data
- Error checking
- Data retransmission when required

Method 1: Using two IDLE windows

Step 1: Create server.py

Open IDLE:

```
import socket

# Create a socket
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

# Bind the socket to IP address and port
server_socket.bind(("localhost", 12345))

# Listen for client connection
server_socket.listen(1)

print("Server is waiting for connection...")

# Accept client connection
client_socket, address = server_socket.accept()

print("Client connected:", address)

# Receive message from client
message = client_socket.recv(1024).decode()

print("Message from client:", message)

# Send response to client
response = "Hello Client! Your message has been received."
client_socket.send(response.encode())

# Close connection
client_socket.close()
server_socket.close()
```

Now press:

F5 → Run Module

You will see in the Python Shell:

```
========== RESTART: C:/Users/COMP/Desktop/Python Practical/Server.py ==========
Server is waiting for connection...
```

Step 2: Open another IDLE window

Save it as:

Client.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client_socket.connect(("localhost", 12345))

message = "Hello Server! My name is Vrushali."

client_socket.send(message.encode())

response = client_socket.recv(1024).decode()

print("Response from server:", response)

client_socket.close()
```

Press:

F5 → Run Module

The client Shell will display:

```
================== RESTART: C:/Users/COMP/Documents/Client.py ==================
Response from server: Hello Client! Your message has been received.
```

At the same time, the **server Shell** will display:

```
========== RESTART: C:/Users/COMP/Desktop/Python Practical/Server.py ==========
Server is waiting for connection...
Client connected: ('127.0.0.1', 51401)
Message from client: Hello Server! My name is Vrushali.
```

Conclusion:

The Python socket program was successfully executed, and basic information was exchanged between the client and server using TCP socket communication.